Coarse Acquisition and Analog Handoff in a 3.2 GHz Ring PLL in 130 nm CMOS

PLL130-AFC Project
Design and verification report

***Abstract*—A five-bit calibration controller places a two-stage ring oscillator within the capture range of a 100 MHz-reference charge-pump PLL. Two switches hold both filter nodes during search. A nominal transistor-level transient at 1.2 V and 25 °C selected code 29 and met the acquisition criteria 4.690 µs after calibration enable. The last 500 ns consumed 4.185 mW. A second run reached fine lock but exceeded its precharge-error limit. A fixed-code control remained below target throughout the 6.5 µs observation. The schematics connect these outcomes to the implemented current bank, charge pump, clamp and controller.**

***Index Terms*—automatic frequency calibration, charge-pump PLL, ring oscillator, coarse tuning, transistor-level simulation**

I. Design Objective

The loop must multiply a 100 MHz reference by 32. Its control voltage can tune the oscillator only within the band selected by the current bank. If that band is too low, the control voltage can approach the supply before the divided clock reaches the reference frequency. The fixed-code experiment in Section III exhibits this behavior.

The starting 180 nm project describes two gated counts, a numeric frequency error E, and a band update $S \leftarrow S + E$ [1]. Its supplied two-stage VCO files informed the ring topology. Its separate four-stage top-level SPICE file instantiates a conventional PLL. The present 130 nm controller adapts the two-phase direction decision and bounded search of Jeong *et al.* [2]. Their LC oscillator uses a capacitor bank. Here the search controls ring current branches, and the two observation phases come from the reference clock and its inverse. Five static CMOS toggle stages divide the clock. Drain-switched mirrors replace the regulated source-switched pump shown in the student report.

II. Signal Path and Handoff

The physical feedback clock closes the analog loop and supplies the AFC measurement input (Fig. 1). During coarse search the pump inputs are disabled. Transmission gates connect CTRL and the midpoint of the filter branch to a common half-supply reservoir. Holding both nodes limits the initial voltage difference across the filter resistor at handoff. Once the controller accepts a band, it retains the code, releases the clamp and permits the PFD pulses to reach the pump.

TABLE I
Frozen counts from the selected calibration

Code	Lead	Lag	Feedback	Decision
16	9	9	7	Slow
24	25	24	22	Slow
28	81	81	79	Slow
30	67	67	69	Fast
29	130	128	129	Near

TABLE II
Nominal closed-loop outcomes

Quantity	AFC enabled	Fixed code 16
Tail frequency (GHz)	3.199994	2.469028
Tail mean CTRL (V)	0.64750	1.18790
Tail supply power (mW)	4.18516	3.38261
Absolute acquisition (µs)	4.98955	none in 6.5
Enable to acquisition (µs)	4.68953	—

III. Acquisition Experiment

Spectre 18.1.0.077 simulated the selected netlist at the typical corner, 25 °C and 1.2 V. The bench used ideal reference and bias sources, initial conditions for the ring and filter, a 4 ps maximum integration step and a 40 ps saved strobe. The deterministic 6.5 µs observation contains 162 501 points. Values below come from the archived runs [3].

The reader interpolated rising zero crossings of $v_0 - v_{180}$. For N edges at times $t_1, \ldots, t_N$ in a 100 ns window it computed $f_{\mathrm{win}} = (N-1)/(t_N - t_1)$. Windows advanced by 50 ns. Acquisition required frequency error within 0.1% of target, unwrapped feedback/reference timing variation within 100 ps peak to peak over a 500 ns hold, and output swing of at least 50 mV peak to peak. Every subsequent complete hold window also had to pass. A separate handoff criterion limited the held CTRL error to 30 mV relative to 0.6 V. The first three decisions increased the coarse code. At code 30 the feedback count exceeded the lead count, so the last step selected 29. DONE rose at 3.850960 µs and FineEnable at 3.870949 µs. The final mean CTRL was 0.64750 V and differential swing was 0.22705 V peak to peak.

With calibration disabled, CTRL approached the positive rail while the output remained below target. This control removed the AFC loading and clamp circuitry. Its tail power describes an uncaptured operating point. In the

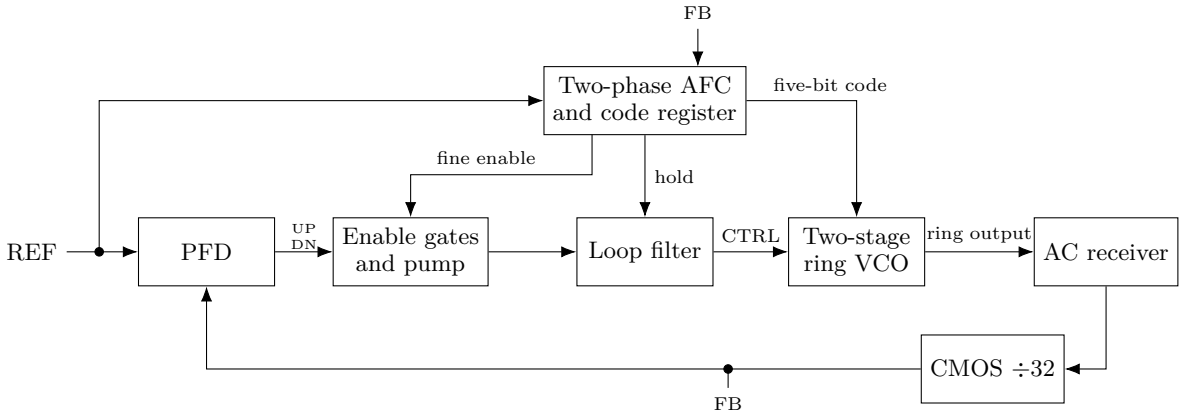


Fig. 1. Selected PLL. Both FB labels identify the physical divided clock. AFC also supplies PFD reset. Each clamp connects one filter node to the half-supply reservoir.

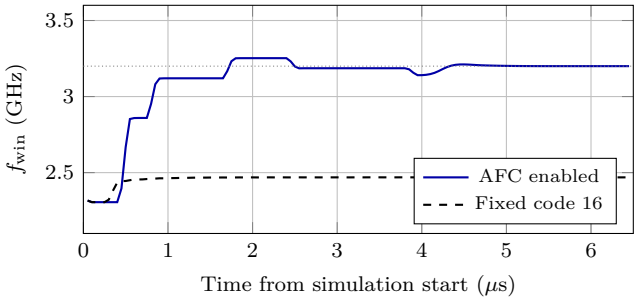


Fig. 2. Output frequency at the centers of all 129 complete windows in each archived trace. The dotted line marks 3.2 GHz.

enabled circuit, supply energy from calibration enable to lock was 24.388 nJ.

The weaker-clamp run reached fine lock but exceeded the 30 mV precharge limit with 35.547 mV maximum error. Increasing the switch widths and hold capacitance reduced the error to 19.313 mV. This comparison measures their combined effect. A separate unloaded VCO run at code 26 with 60 sidebands gave −73.478 and −95.231 dBc/Hz at 1 and 10 MHz offsets. Integrated closed-loop phase noise, RMS jitter, reference spurs and post-layout acquisition remain unverified.

IV. Using the Sources

The canonical AFC-on bench, fixed-code control, measurement ledgers and window series are available in the repository [3]. The Spectre decks define the electrical experiment. OCEAN scripts export noise data from stored results. Project SKILL includes a one-line Library Manager launcher. A local installation supplies the process models. The source archive distinguishes the original simulation input hashes from portable copy hashes.

V. Transistor Schematics of the Selected Loop

The following schematics follow the instantiated R1 strong-clamp hierarchy. The source identifiers in Table III can be searched verbatim in the accompanying netlists. A repeated net label denotes electrical continuity within its stated module. An NMOS bulk connects to its local negative rail and a PMOS bulk to its local positive rail. The captions identify RF devices. Each of these symbols represents a four-terminal RF device wrapper at the design boundary. Its finger count belongs to one instance. All component values refer to the selected pre-layout bench.

A. Oscillator and band selection

The oscillator closes two differential stages with one polarity swap. Stage 0 takes (o_0, o_{180}) and drives (o_{90}, o_{270}). Stage 1 takes (o_{270}, o_{90}) and drives (o_0, o_{180}). In each stage the input pair and the regenerative pair have separate tail nodes (Fig. 3). The control branches act on the input-pair tail only (Fig. 4). The selected sizing is given by the actual instance parameters. In particular the RF input devices have twelve fingers of width 3 μm each.

B. Clock conversion and frequency division

The receiver couples the positive differential-pair output to a CMOS clock restorer (Fig. 6). Feedback biases the first inverter near its switching threshold. Five toggle stages then divide the recovered clock by 32 (Fig. 7).

C. Phase detector, charge pump, and precharge

The two six-device PFD slices receive REF and divider feedback (Fig. 8). After the controller enables fine tuning, two AND gates pass UP and DN to the charge pump. The pump switches the drains of continuously biased mirror devices (Fig. 9). During calibration a transmission gate at each filter node connects that node to the half-supply reservoir (Fig. 10).

VI. Transistor Implementation of the AFC

The R1 AFC compares edges of the 100-MHz reference and the physical divide-by-32 feedback. A higher five-bit code increases the ring frequency. The controller starts at code 16 and can test four further candidates using steps of

TABLE III
SCHEMATIC-TO-NETLIST MAP FOR THE SELECTED LOOP.

Selected instance	Netlist module or devices	Function
`XREF`, `XOSC`	`current_ref, two_stage_ring_vco, two_stage_ring`	Bias and two quadrature cells
`XRX`	`pll130_receiver_ac`	Differential input to logic clock
`XD`	`pll130_div32_cmos, pll130_div2, pll130_dff, pll130_latch`	Five toggle stages
`XPFD`	`pll130_pfd, pll130_pfd_tspc`	Reference and feedback phase comparison
`XCPUPEN`, `XCPDNEN`, `XCP`	`pll130_and2, pll130_cp_drain, pll130_cpd_inv`	Pulse gating and charge transfer
Top-level passives and switches	`CSHUNT, RSERIES, CSERIES, CHOLD, MHOLD*, PHOLD*`	Integration and precharge

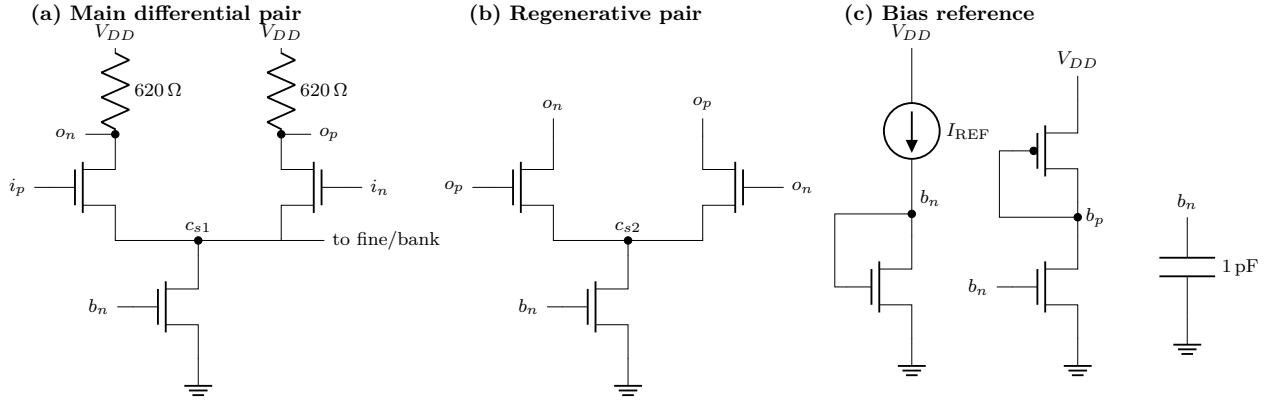


Fig. 3. One ring cell and the common bias source. The six devices in (a,b) use RF wrappers. Input, regenerative, main-tail and regenerative-tail finger counts are 12, 8, 12 and 16, respectively. Each finger has $W = 3\,\mu$m. Pair devices have $L = 0.13\,\mu$m and tail devices $L = 0.39\,\mu$m. The bias source supplies $13.5594\,\mu$A. The two bias NMOS devices have $W/L = 8/3.12$ and the PMOS has $W/L = 4.06783/0.39$, in μm.

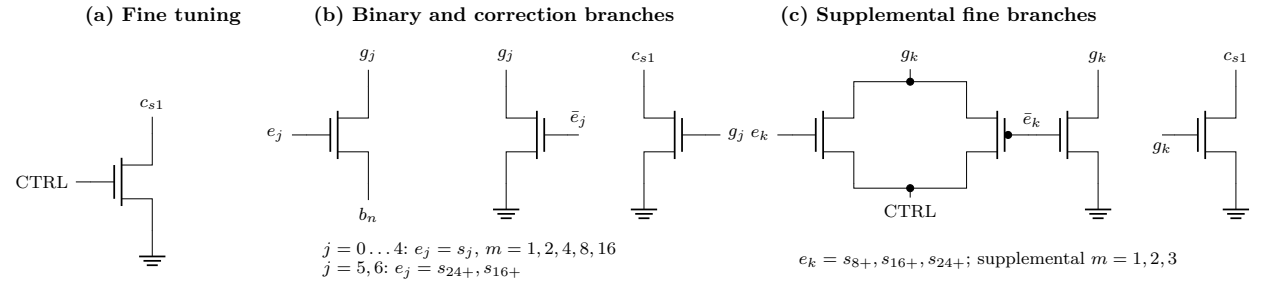


Fig. 4. All three tuning paths in one cell. The fine device has $W/L = 0.8/5.2\,\mu$m. Binary sinks have unit $W/L = 1.1/0.39$. The two correction widths are 0.47143 and 0.23571 μm for 24+ and 16+, respectively. For (c), the selected device gate receives CTRL through a transmission gate and the disabled gate is grounded. Supplemental devices have $W/L = 0.2/5.2$. The decoder forms $s_{8+} = s_3 \vee s_4$, $s_{16+} = s_4$ through two inverters and $s_{24+} = s_3 \wedge s_4$. The `s_to_d` buffers in Fig. 5 derive both polarities. The labels g_j and g_k refer to separate branch families.

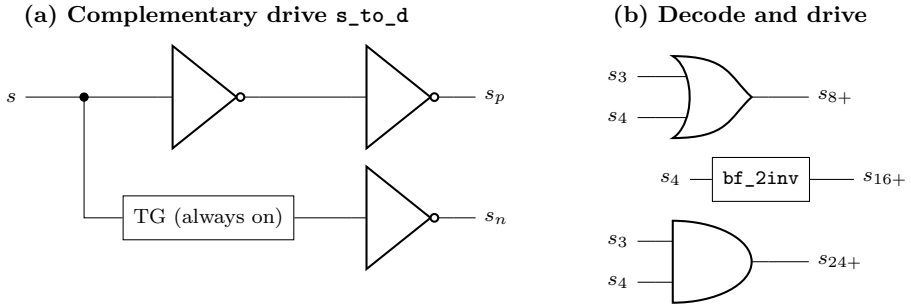


Fig. 5. VCO bank logic. Each cell contains eight complementary drivers, one per binary bit and one per decoded threshold. The always-on TG connects the input to a single inverter with its NMOS gate at VDD and PMOS gate at VSS. The other branch uses two inverters. The `bf_2inv` buffer is also a pair of inverters in series. OR2 uses NOR2 followed by an inverter, and AND2 uses NAND2 followed by an inverter. All VCO logic inverter devices have $W_n/W_p = 0.5/1.8\,\mu$m and $L = 0.13\,\mu$m.

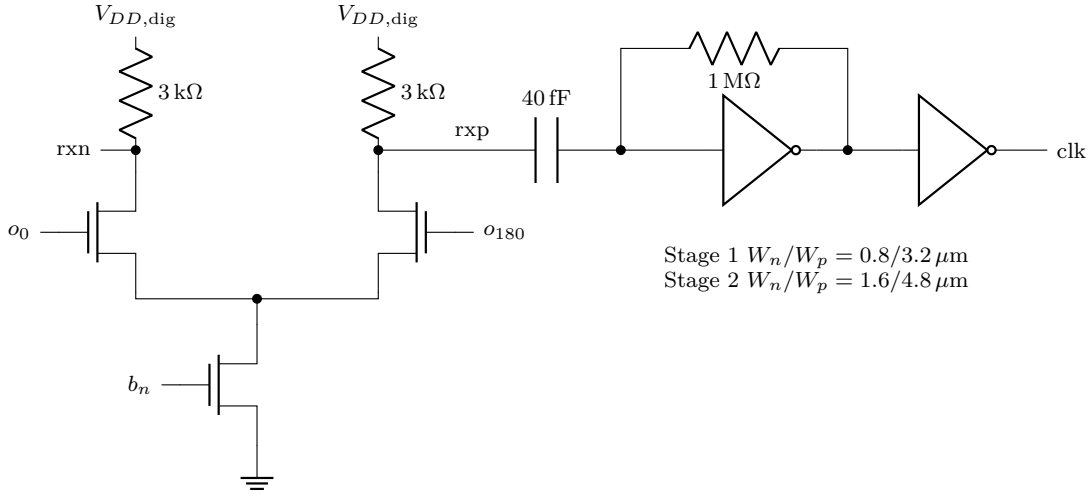


Fig. 6. Selected AC-coupled receiver. Each differential input uses four RF fingers of 1.5 μm at $L = 0.13\,\mu$m. The tail uses 12 fingers of 2.5 μm at $L = 0.39\,\mu$m. The bench sets the coupling capacitance to 40 fF.

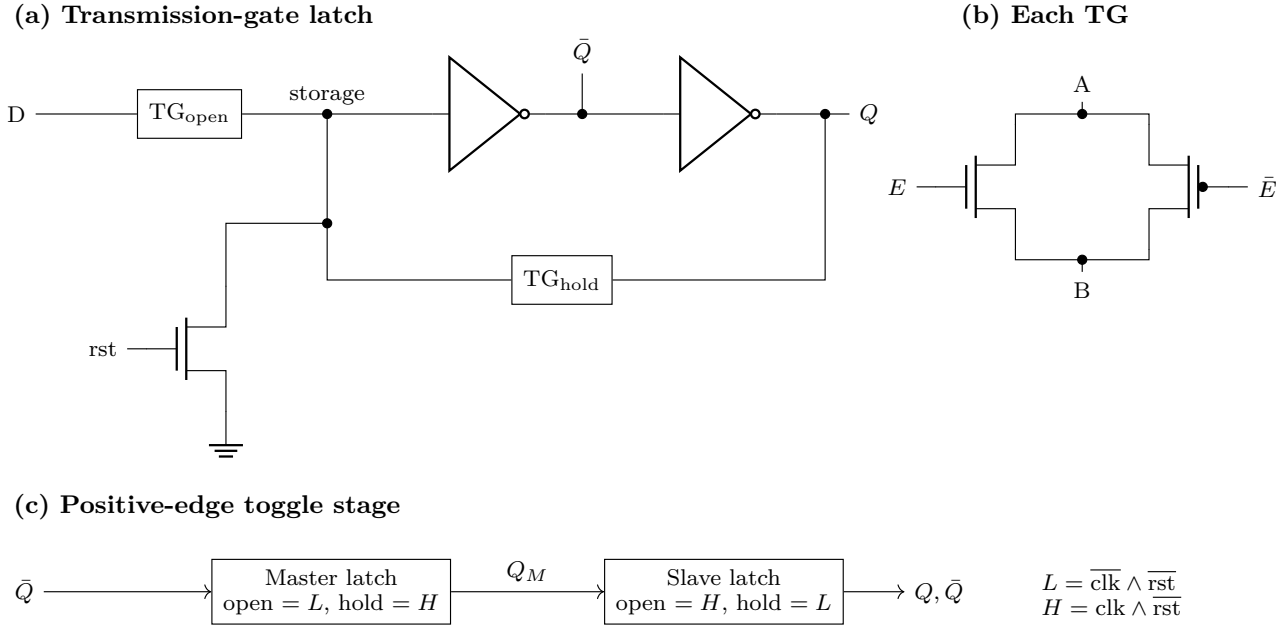


Fig. 7. Physical divider hierarchy. Five copies of (c) cascade through their Q outputs, giving q_1, q_2, q_3, q_4, fb. A 5-fF capacitor loads fb. Each latch has two TGs, two inverters and one reset NMOS. TGs have $W_n/W_p = 1.2/3.6\,\mu$m, latch inverters 0.5/1.5 μm and the reset NMOS $W = 2\,\mu$m. Inverters outside the latch generate complementary TG controls.

8, 4, 2, and 1. Each candidate reaches the detector before the controller can hand it to the fine loop.

The first admitted FB edge samples the reference level and assigns the two reference phases to lead and lag counters (Fig. 11). The lead counter starts with one credit. A chain of 16 inverters guards reference-counter reset release after that first edge. The controller samples each Gray word through two flip-flops per bit, then decodes it. With lead, lag, and divided-feedback counts C_L, C_G, C_D, respectively, it qualifies a slow decision when $C_G - C_D \geq 2$ and a fast decision when $C_D - C_L \geq 2$. The unsigned subtractors check the carry first. Two consecutive observations in the same direction qualify a request. Reaching lag count 128 without a qualified direction ends a trial as near.

An empty ARM window times out after 32 reference clocks. MEASURE has a 192-clock watchdog. The circuit stores the three decision words and direction before ending the measurement. On success, two further reference clocks separate DONE from FineEnable. HOLD is the complement of FineEnable. The PFD reset is asserted when reset is high or FineEnable is low.

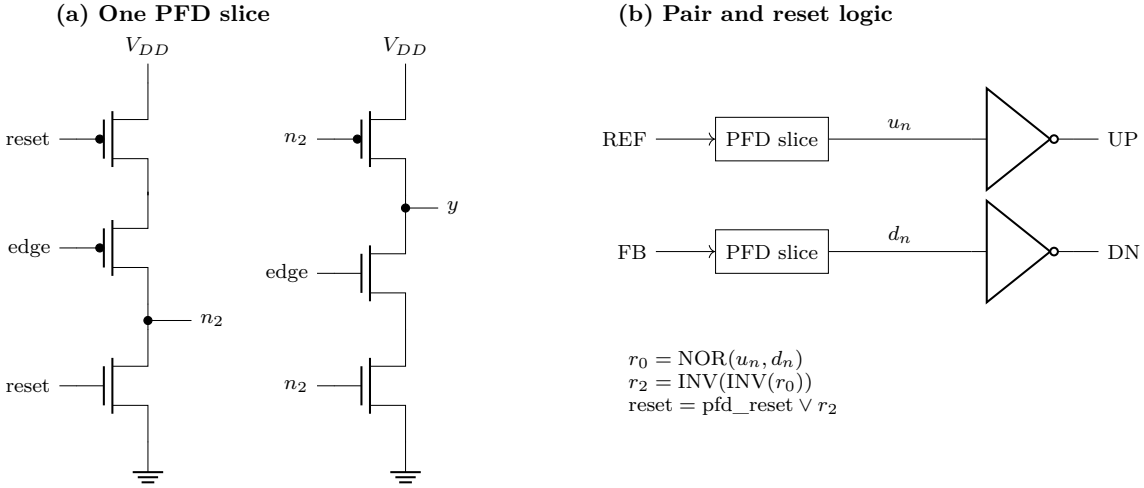


Fig. 8. Six-MOS single-phase dynamic slice (left) and PFD assembly (right). The two PMOS devices in the input column are stacked. The reset NMOS discharges n_2. In the output column n_2 drives the PMOS and bottom NMOS, while the edge input drives the middle NMOS. The two explicit inverters preserve the delays of the reset path.

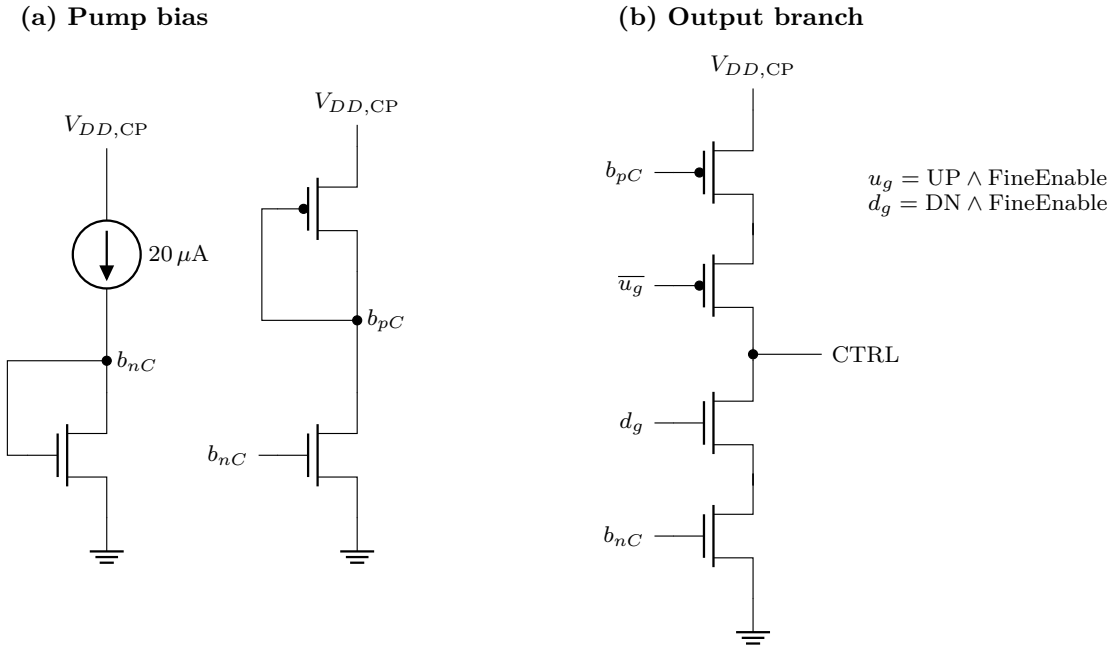


Fig. 9. Drain-switched charge pump. Bias mirrors have $W_n/L = 4/1.04$ and $W_p/L = 12/1.04$ in μm. Output switches have $W_n/L = 2/0.13$ and $W_p/L = 6/0.13$. A CMOS inverter generates $\overline{u_g}$. The separate oscillator bias of Fig. 3 has its own bias nodes and reference current.

A. Static CMOS Cells

The generator expands the gates into one flat Spectre subcircuit. Figure 13 shows the transistor connections of the gate constructors. All devices use $L = 0.13$ μm. NMOS bulk connects to VSS and PMOS bulk connects to VDD. Repeated input labels in a panel denote the same electrical net.

The default inverter has $W_n/W_p = 0.8/2.4$ μm. Its latch and clock-driver variants use 0.5/1.5 and 8/24 μm. NAND2, NOR2 and NAND4 use per-device widths of 1.6/2.4, 0.8/4.8 and 3.2/2.4 μm, respectively. The TG uses 1.2/3.6 μm. For an XOR, the generator forms $U = \overline{AB}$ and then $Y = \overline{\overline{AU}\,\overline{BU}}$. A two-input multiplexer uses NAND2 cells and a complemented select. These same cells form the Gray converters, incrementers, difference detectors and code update logic.

Two latches form one DFF. The preset variant inverts D and exposes the slave's complementary output. Each shared clock group forms K_d with six cascaded inverters. With active-high reset R, the interlock computes

$$L = \overline{K}\,\overline{K_d}\,\overline{H}\,\overline{R}, \qquad H = K\,K_d\,\overline{L}\,\overline{R}. \tag{1}$$

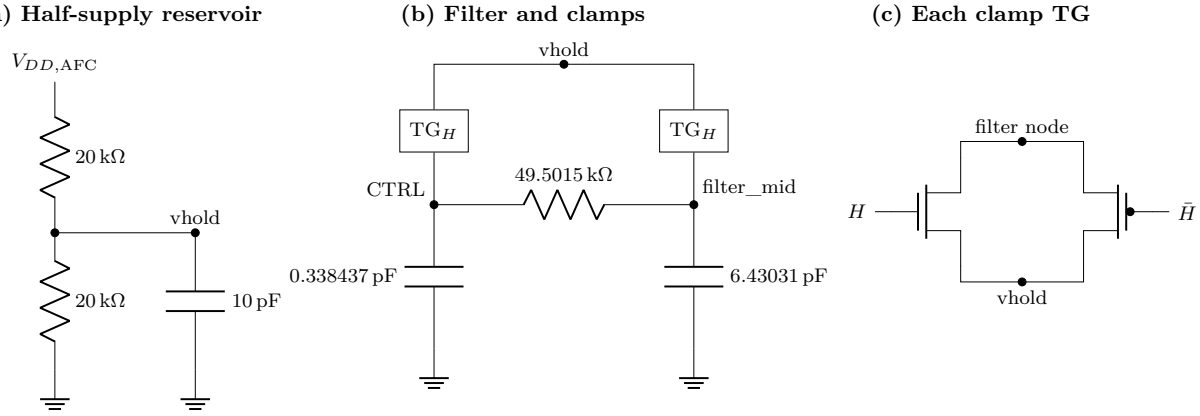


Fig. 10. Both filter nodes precharge from the same passive reference during coarse calibration. The complementary control pair is H = hold_all and $\bar{H} = \mathrm{INV}(H)$. Each physical transmission gate uses a 16/0.13 μm NMOS and a 32/0.13 μm PMOS. These ratios identify the strong-clamp revision.

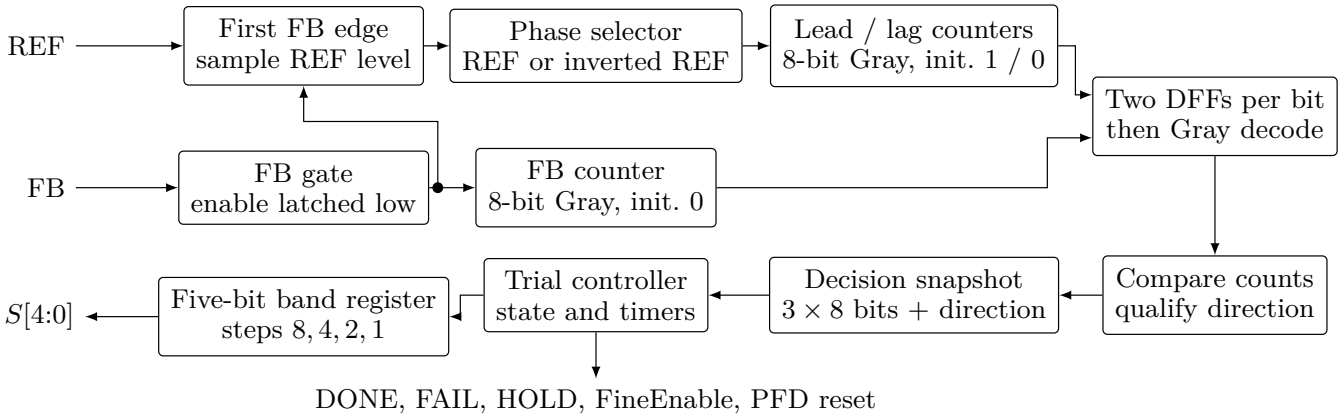


Fig. 11. R1 frequency detector and search controller. The phase selector also receives REF and its complement. The gated FB clocks both the FB counter and the first-edge sample. The synchronized counts share the controller clock. Trial control supplies the counter enables and resets.

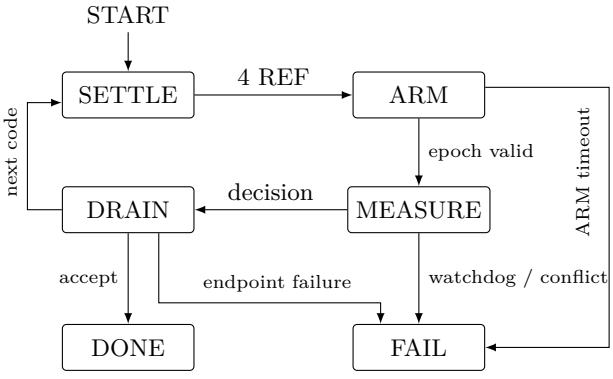


Fig. 12. R1 state transitions. DRAIN holds the captured counts for two controller cycles. A terminal decision accepts a near result or the last tested candidate. An outward direction at code 1 or 31 on a terminal decision sets FAIL.

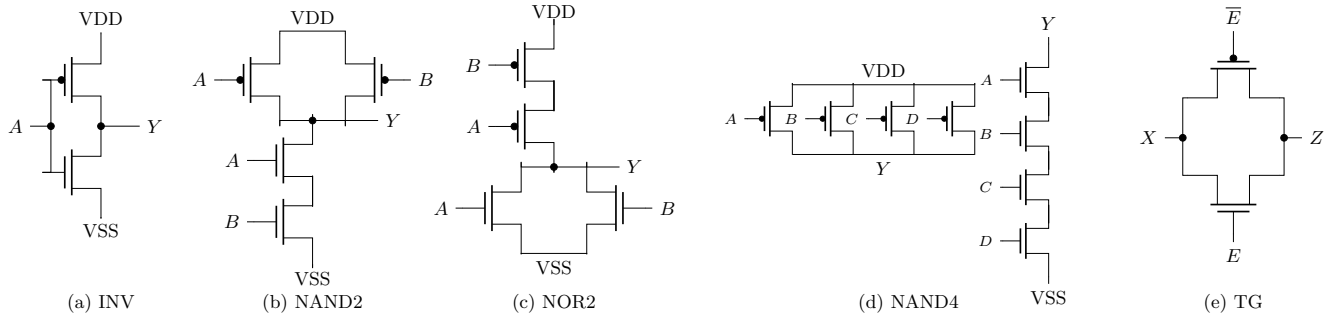
A NAND4 and two large inverters drive each phase and its complement.

R1 contains 146 DFFs and one additional latch on the feedback clock gate. The 293 latches use 586 TGs, 586 latch inverters and 293 reset NMOS devices. Outside these latches, the generator emits 531 inverters (including clock drivers), 740 NAND2s, 79 NOR2s and 20 NAND4s. The total is $293(9) + 531(2) + 740(4) + 79(4) + 20(8) = 7{,}135$ MOS instances in the AFC core. These counts can be checked against the selected R1 gate generator, its port manifest and the generated transistor netlist.

VII. Historical Circuit Variants

Two early receiver families share a resistively loaded differential pair. The load experiment exposes the two drain voltages directly. The direct-coupled clock receiver adds two CMOS inverters at the positive output (Fig. 15). In the selected receiver, a coupling capacitor and a feedback resistor set the first inverter's input bias (Fig. 6). This distinction matters when comparing the measured loading and clock waveforms.

The original AFC core R0 contains 6,908 MOS instances and 145 DFFs. Its FB counter uses the physical FB clock directly. Reset follows RST$\vee\overline{\mathrm{count_run}}$. R1 adds a latched



(a) INV (b) NAND2 (c) NOR2 (d) NAND4 (e) TG

Fig. 13. Distinct gate topologies of the R1 AFC. For the NAND4 the two networks share output Y. The transmission gate (TG) connects X and Z when $E = 1$. The latch adds the reset NMOS shown in Fig. 14.

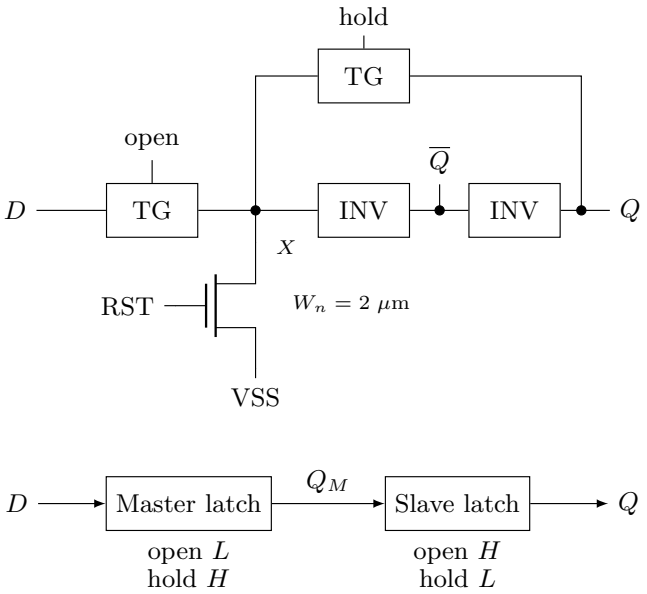


Fig. 14. R1 resettable static latch (top) and ordinary positive-edge DFF (bottom). Each TG receives an enable and its complement. The feedback TG returns Q to storage node X during hold. Active reset pulls X to VSS in each latch.

FB-clock enable and holds counter reset deasserted during DRAIN, as shown in Fig. 11. The archive also records the R2/counter9 proposal as a prepared design option. The nominal closed-loop result in this report belongs to R1.

The simulator additionally accepts eight project-authored behavioral modules. They represent a charge pump, precharge path, linear VCO, divide-by-32 counter, PFD, search controller, handoff controller and AFC unit-test source. Their equations and event handlers provide stimuli and functional substitutes in staged benches. The transistor counts in the selected design apply to the MOS implementations.

The companion `module-index.md` maps all 32 distinct Spectre subcircuit names to these drawings and identifies the eight Verilog-A modules by name.

VIII. Conclusion

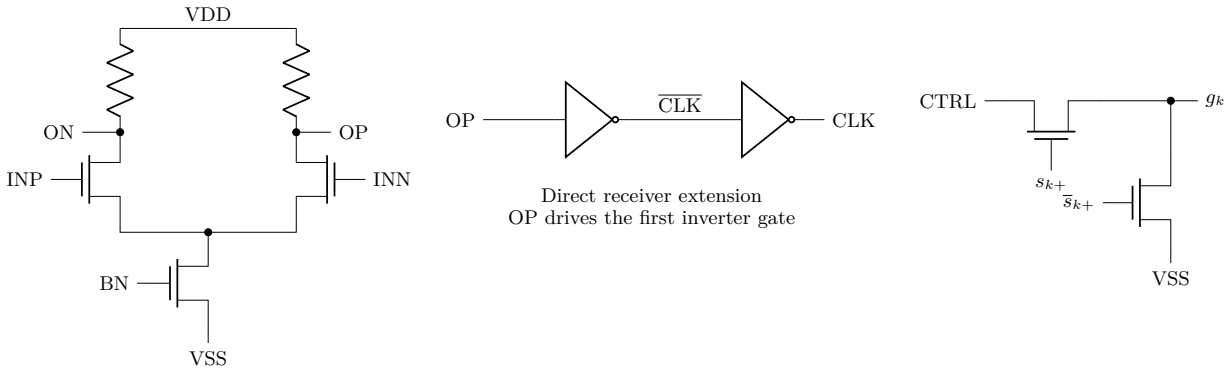
The selected controller reached code 29 and acquired sustained nominal lock 1.119 μs after fine enable. The weaker-clamp run exceeded its precharge tolerance even though the fine loop acquired. The fixed-code loop exhausted its analog tuning range within the observed interval.

References

[1] *A 3.2 GHz Fast-Locking and Low-Jitter PLL with Automatic Frequency Calibration Technique in 0.18-µm CMOS Technology*, supplied undergraduate design report, Figs. 1, 2, 4, 7, 9, 11 and 12.

[2] C.-Y. Jeong, D.-H. Choi, and C. Yoo, *A Fast Automatic Frequency Calibration (AFC) Scheme for Phase-Locked Loop (PLL) Frequency Synthesizer*, Proc. IEEE Radio Frequency Integrated Circuits Symp., 2009, pp. 583–586.

[3] PLL130-AFC Project, *Nominal Transient and Acquisition Evidence*, Oct. 2026. https://github.com/handsomegary/pll130-afc.



(a) Shared CML front end (b) Two added CMOS inverters (c) Early fine-bank gate drive

Fig. 15. Archived load and oscillator variants. With 1-kΩ loads the CML-only cell has four input fingers of either 1.5 or 3 μm per device. The direct receiver defaults to 3-kΩ loads and adds (b). Both use a 12-finger tail with 2.5-μm fingers. In (c), each of $k = 8, 16, 24$ uses an NMOS to pass CTRL to a supplemental sink gate and a second NMOS to discharge it. The later ring adds a parallel PMOS on the CTRL path.

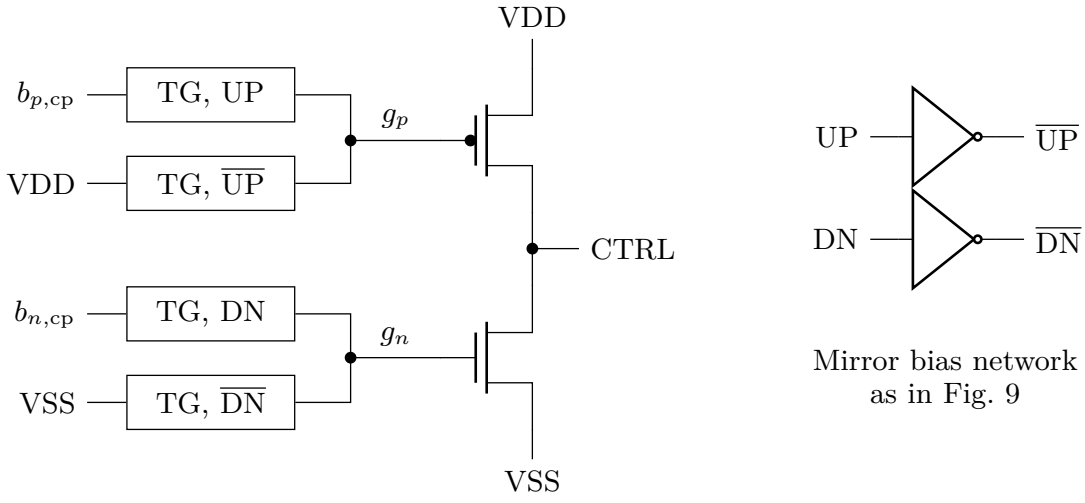


Fig. 16. Gate-switched charge-pump candidate. Each of the four TGs uses an NMOS/PMOS pair with complementary gate drives. The label gives the NMOS enable. With UP high the PMOS output gate selects the mirror voltage, and with UP low it selects VDD. DN likewise selects the NMOS mirror voltage or VSS. The TG and inverter widths are 0.5/1.5 μm at $L = 0.13$ μm. The NMOS and PMOS output widths are 4 and 12 μm at $L = 1.04$ μm.